Zero-gravity laboratory Specification

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Revision	A
Status	APPROVED
Classification	INTERNAL
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1. Purpose

This specification defines the performance, interface and verification requirements for the zero-gravity laboratory (RL-ZGL) of the Research Laboratories on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-RL-04 (WBS 1.16.4), owned by Quinn Aster.

The prime supplier for flight hardware is Fabritech. Responsible engineer: Talia Orlen.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-RL-DWG-0203	Zero-gravity laboratory general arrangement drawing
SHV-RL-BOM-0303	Zero-gravity laboratory bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-RL	Interface Control Document: Power Distribution / Research Laboratories
SHV-ICD-RL-CN	Interface Control Document: Research Laboratories / Computer Core & Data Network

4. Description

The research laboratories host materials science, biology and astronomy groups: wet and dry labs, cleanroom suites, a zero-gravity laboratory and the station observatory.

The zero-gravity laboratory is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
RL-ZGL-R01	The zero-gravity laboratory shall provide a lab volume of 5,025 m³.	5,025 m³	±5%	Inspection

5.1 General requirements

ID	Requirement	Verification
RL-ZGL-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
RL-ZGL-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
RL-ZGL-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
RL-ZGL-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
RL-ZGL-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the RL test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-01-06	Initial release.	Talia Orlen